

Muralidharan Vivekananthan

Backend Software Engineer | Go • AWS • PostgreSQL • REST APIs

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Summary

Backend-focused Software Engineer with nearly 3 years of experience building API integrations, automation workflows, and cloud-based backend solutions for production SaaS environments. Hands-on experience with Go, AWS, PostgreSQL, REST APIs, asynchronous processing, and data synchronization. Experienced in breaking down ambiguous requirements, implementing backend solutions within existing systems, troubleshooting production issues, and improving system performance and reliability.

Skills

🔗 **Languages:** Go, JavaScript, SQL

🔗 **Backend:** REST APIs, API Integrations, Webhooks, Event-Driven Architecture, Backend Automation

🔗 **Databases:** PostgreSQL

🔗 **Frontend:** React

🔗 **AWS:** S3, SQS, Lambda, ECS, API Gateway, EventBridge, EC2, RDS, Parameter Store

🔗 **DevOps:** Docker, GitHub Actions, Terraform/Terragrunt, CI/CD, AWS CLI

🔗 **Tools:** Git, Postman, n8n

🔗 **Engineering:** System Architecture, API Design, Asynchronous Processing, Message Queues, Fault Handling, Root Cause Analysis, Log Analysis, Debugging

🔗 **Other:** Zoho Products, Deluge Scripting, VBA Excel

Projects

ApplyFlow — Event-Driven Job Application Automation

[Go • React • PostgreSQL • AWS RDS • S3 • SQS • EventBridge • EC2 • ALB • Docker • GitHub Actions • Gmail API]

- Designed and built an event-driven job outreach platform that accepts job-application campaigns and asynchronously sends personalized emails through the Gmail API.
- Built the backend in Go with PostgreSQL hosted on Amazon RDS for persistent users, campaigns, templates, outreach state, and progress tracking.
- Implemented XLSX upload and validation, duplicate-recipient prevention, configurable email templates, resume attachment handling, campaign scheduling, cancellation, and progress tracking.
- Designed asynchronous email processing using Amazon SQS and a separate worker service, decoupling API requests from email delivery and supporting failure recovery and at-least-once message processing.
- Implemented Google OAuth authorization with authorization-code exchange and access/refresh token handling for authenticated Gmail API access.
- Used Amazon S3 for resume and spreadsheet storage and EventBridge-based scheduling for time-triggered campaign processing.
- Built the React frontend and configured the AWS deployment using Docker, RDS, EC2, Application Load Balancer, and security groups.
- Implemented campaign state tracking and progress updates across individual outreach records, allowing users to monitor sent, failed, and in-progress applications from the dashboard.
- Designed campaign cancellation handling to stop pending scheduled outreach and prevent further processing of cancelled campaigns.

Experience

KAVA CRM

Remote - Australia

CRM Developer — Backend / Integrations

February 2025 – Present

- Worked on existing backend integration workflows in **Go**, connecting Zoho CRM with Australian insurance policy systems, including Insight and Winbeat, and Microsoft SharePoint.
- Implemented enhancements to data synchronization workflows that retrieve data from external APIs and databases, transform and validate records, and push processed data into Zoho CRM.
- Optimized Insight/Winbeat synchronization processing, reducing execution time from approximately **20 minutes to under 5 minutes**.
- Reworked existing SharePoint integration processing in **Go**, reducing execution time from several minutes to **under 1 minute**.
- Extended existing integration workflows to support multiple policy systems through configuration-driven processing and AWS S3 CSV-based integrations.
- Worked with **AWS Lambda, ECS, S3, SQS, API Gateway, EventBridge, EC2, and Parameter Store** across integration, scheduling, and deployment workflows.
- Created and maintained a spreadsheet comparison and reconciliation tool to validate synchronized data and reduce manual verification effort.
- Implemented changes to deployment configuration and automation workflows, reducing credential and configuration setup time from approximately **15 minutes to a few seconds**.
- Used third-party **Terraform/Terragrunt** infrastructure templates for customer deployments and configuration, gaining practical experience with infrastructure-as-code workflows.
- Investigated production integration failures using logs, API responses, data comparisons, and workflow execution details to identify and resolve synchronization and processing issues.

ZUPER

Chennai, India

Software Engineer Trainee

October 2023 – December 2024

- Developed backend automation workflows and API integrations connecting Zuper with SaaS platforms including Zoho CRM, Zoho Books, Zoho Desk, Zoho Inventory, Zoho Projects, Freshsales, Freshdesk, Zendesk, and HubSpot.
- Built and maintained API integrations and automation workflows using Deluge, Node.js, Express.js, and n8n.
- Translated client business requirements into practical integration and automation workflows across interconnected SaaS systems.
- Implemented data synchronization workflows to automate information flow between business applications and reduce manual processing.
- Investigated and resolved API, data synchronization, and automation failures across customer environments through log analysis and troubleshooting.
- Developed custom UI functionality within Zuper to support customer-specific requirements.

Education

PSNACET

Bachelor of Engineering • 8

Computer Science

Dindigul, India • Aug 2019 - June 2023